

Scientific Enquiry in Biology

Science

May, 2026

Scientific Enquiry in Biology (A07461)

Short Title: Scientific Enquiry in Biology
Faculty Science and Computing
Department Science
Cluster Biology
Author KTAMBLING
Credits 5

Level: Introductory

Description of Module / Aims

This module is comprised of a series of in-class discussions, debates and presentations on current topical issues in biology, as well as lectures on specific topics. The aim of this module is to expose students to current research and topics of relevance in biology, and to aid in the development of skills in reviewing scientific literature by researching published sources of information.

Programmes

SCIE-0056	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	2 / 3 / M
SCIE-0056	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	2 / 3 / M

Indicative Content

- The fundamentals of scientific research and how research is conducted
- How research findings are disseminated via publications in peer-reviewed scientific journals etc.
- How to search and read scientific research papers
- How to deliver an effective presentation
- Media reports on topical issues in biology including, safety of vaccines, genetically modified organisms, biofuels, cloning, antibiotic-resistant bacteria, emerging pathogens and topical pathogens such as Ebola virus, diseases and disorders, stem cell research, obesity, allergy, functional foods, new drugs, synthetic meat, probiotics, the gut microflora etc.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Discuss scientific research papers and media reports on topical biology-related issues.
2. Use appropriate tools for gathering scientific information, e.g. scientific databases, library etc.
3. Operate effectively as a member of a team.
4. Communicate scientific principles to peers.

Learning and Teaching Methods

- Lectures.
- Self-directed learning.
- In-class discussions, debates, presentations.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Assignment</i>	45%	1,2
<i>Presentation</i>	55%	3,4

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. No attempt to research, discuss or debate issues.

40%-49%: Demonstrate a limited knowledge of biology. Some participation in debate but little evidence of independent thinking.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Participate actively in debate, communicating clearly with some independent ideas.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- "Food Safety Authority of Ireland." Food Safety Authority of Ireland. www.fsai.ie
- "Moodle used to provide links to useful websites." <https://moodle.wit.ie/login/index.php>
- "PubMed." www.ncbi.nlm.nih.gov/pubmed
- "Science Direct." www.sciencedirect.com
- "TED talks." www.ted.com/talks

Requested Resources

- Room Type: Computer Lab